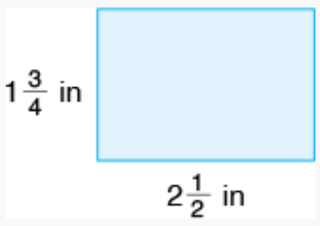
Problems with Mixed Numbers

When you are multiplying or dividing, you should always change mixed numbers to improper fractions.

Example 1

Hide

A rectangle measures $1\frac{3}{4}$ inches by $2\frac{1}{2}$ inches. What is the area of the rectangle in square inches? (***Hint:*** To find the area of a rectangle, multiply the length by the width.)



Area = length x width

$1\frac{3}{4} = \frac{4\times1+3}{4} = \frac{7}{4}$ | $2\frac{1}{2} = \frac{2\times2+1}{2} = \frac{5}{2}$

Step 1

Change mixed numbers to improper fractions.

$1\frac{3}{4} = \frac{7}{4}$ $2\frac{1}{2} = \frac{5}{2}$

Step 2

Multiply the improper fractions.

$\frac{7}{4} \times \frac{5}{2} = \frac{35}{8}$

Step 3

Change the improper fraction to a mixed number.

8x4=32 (3 left over)

$\frac{35}{8} = 4\frac{3}{8}$

The area of the rectangle is $\mathbf{4\frac{3}{8}}$ **square inches.**

As you work, remember that your final answer should be in lowest terms. Always check the fraction part of the answer to see if it can be simplified.

Example 2

Hide

Bill used $2\frac{1}{2}$ gallons of paint to cover one side of a warehouse. Each side is the same size. How many sides could he paint if he had $6\frac{1}{4}$ gallons of paint?

Step 1

Change mixed numbers to fractions.

$6\frac{1}{4} \div 2\frac{1}{2} = \frac{25}{4} \div \frac{5}{2}$

$6\frac{1}{4} = \frac{4\times6+1}{4} = \frac{25}{4}$ | $2\frac{1}{2} = \frac{2\times2+1}{2} = \frac{5}{2}$

Step 2

Invert the divisor and change the ÷ to a ×.

$\frac{25}{4} \times \frac{2}{5}$

Step 3

Cancel if possible and multiply.

$\frac{\cancel{25}^{5}}{\cancel{4}_{2}} \times \frac{\cancel{2}^{1}}{\cancel{5}_{1}} = \frac{5\times1}{2\times1} = \frac{5}{2}$

Step 4

Change improper fractions to mixed numbers.

$\frac{5}{2} = 2\frac{1}{2}$

Bill could paint $\mathbf{2\frac{1}{2}}$ **sides** with $6\frac{1}{4}$ gallons of paint.

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Lesson 5: Operations with Fractions

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